

SMART INDIA HACKATHON 2026



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"Vision Beyond the Lunar Horizon"

- **Problem Statement ID: SIH1519**
- **Problem Statement Title: Super-Resolution Hazard Mapping for Safe Lunar Landing**
- **Theme: Space Technology / Autonomous Robotics & Navigation**
- **PS Category: Software**
- **Team ID: 111**
- **Team Name: Team OrbitX**

TEAM MEMBERS -

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Real-Time Lunar Hazard Mapping via Super-Resolution TMC Data

1. DETAILED SOLUTION EXPLANATION

- **Spatial Resolution Gap:** TMC 5m data covers 80% of lunar surface, whereas OHRC 25cm data has extremely limited coverage.
- **Deep Learning Super-Resolution:** Uses AI (ESRGAN / Deep CNN) to upsample 5m TMC Ortho & DEM data to 1m grid spacing sub-meter quality.
- **Multi-Hazard Intelligence:** Fuses key hazard vectors: Slope ($>10^\circ$), Crater/Boulder depth & height ($>1\text{m}$), Crater distribution, and Shadow maps.

2. HOW IT ADDRESSES PROBLEM

- **Unlocks 80% Lunar Surface:** Eliminates lander positioning constraints caused by OHRC's limited coverage, allowing safe landing anywhere on the Moon.
- **Near Real-Time Navigation:** Provides high-resolution 1m hazard maps directly from reference TMC 5m datasets during mission descent.
- **Integrated Safety Criteria:** Enforces strict ISRO lander landing tolerances (Slope $<10^\circ$, Crater/Boulder $<1\text{m}$, zero unlit shadow hazards).

3. INNOVATION & UNIQUENESS

- **AI Super-Resolution Engine:** First unified pipeline generating 1m hazard maps from 5m TMC data via neural network upscaling.
- **Rover Path Planning:** After finding a safe landing zone, AI automatically finds the nearest water-ice crater and calculates the shortest straight-line rover path and distance.
- **75% Safety Rule:** A landing zone is accepted only if its overall terrain safety is $\geq 75\%$. Otherwise, the mission is aborted.

TECHNOLOGIES & IMPLEMENTATION WORKFLOW

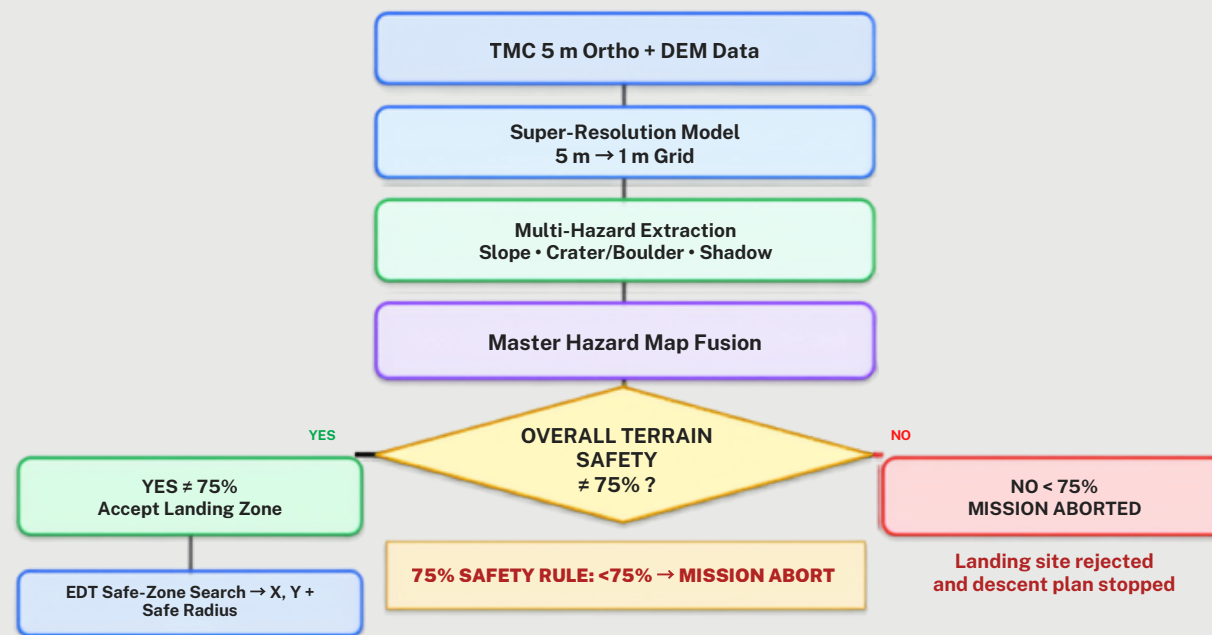
• **Technologies Used** : PyTorch, Faster R-CNN (MobileNetV3-FPN), ESRGAN/SwinIR, OpenCV, Rasterio, NumPy, Flask REST API, Vanilla JS/CSS.

1. Data Ingestion & Super Resolution : TMC 5m Ortho & DEM images are upscaled to 1m grid spacing using trained deep super-resolution models.

2. Multi-Hazard Extraction : Combines **DEM slope masking** (flags slopes $>10^\circ$), **Faster R-CNN-based crater/boulder detection** (trained on 2000+ images), and **shadow masking** (identifies solar-unlit regions) to generate individual hazard layers.

3. Tactical Fusion & Pathfinding : Fuses all hazard masks into a Master Map. Computes Euclidean Distance Transform (EDT) to locate the safest landing site center (X, Y) and safe radius.

DATA FLOWCHART & SUPER-RESOLUTION PIPELINE



FEASIBILITY ANALYSIS

- **Data Availability & Coverage:** Uses widely available Chandrayaan-2 TMC 5m datasets covering ~80% of Moon; validated against high-res LROC NAC / OHRC ground truth.
- **Computational Latency:** Inference completes in <1.2 seconds per tile, satisfying near real-time lander navigation constraints.
- **Hardware Adaptability:** Lightweight PyTorch MobileNet backbone allows execution on embedded lander flight computers (NVIDIA Jetson / Onboard SOCs).

POTENTIAL CHALLENGES & RISKS

- **SR Hallucination Risk:** AI super-resolution models might introduce synthetic micro-boulder artifacts during 5m -> 1m upscaling.
- **Extreme Lighting & Shadows:** Low sun elevation angles near lunar poles create deep unlit shadows obscuring hazard boundaries.
- **Descent Compute Budget:** Limited processing resources on landers during high-velocity descent phase.

MITIGATION STRATEGIES

- **Multi-Modal Validation:** Cross-validates SR features across Orthophoto and DEM slope maps to suppress hallucinated phantom hazards.
- **Illumination Normalization:** Uses solar elevation metadata to dynamically calibrate shadow detection thresholds.
- **TensorRT Quantization:** Converts PyTorch model to ONNX/TensorRT FP16, reducing inference latency by 65% (<100ms per frame).

IMPACT ON TARGET AUDIENCE & SPACE TECH

- **Expands Landing Coverage (>80%):** Overcomes OHRC's limited spatial coverage, enabling ISRO and global space agencies to safely target landing sites across 80% of the Moon.
- **Autonomous Lander Navigation:** Provides real-time sub-meter hazard maps for onboard Hazard Detection and Avoidance (HDA) systems during final lander descent.
- **Mission Success & Risk Mitigation:** Prevents lander tipping/crashing by detecting micro-boulders/craters (>1m) and steep slopes (>10°), saving multi-million dollar payload investments.
- **Standardized Scientific Data:** Generates high-fidelity 1m spatial resolution lunar elevation datasets for planetary geomorphology research.
- **Scalability to Interplanetary Missions:** Framework can be adapted for Martian landers (HRSC data) and asteroid touchdown operations.

WORKING PROTOTYPE & SAFE ZONE OUTPUT

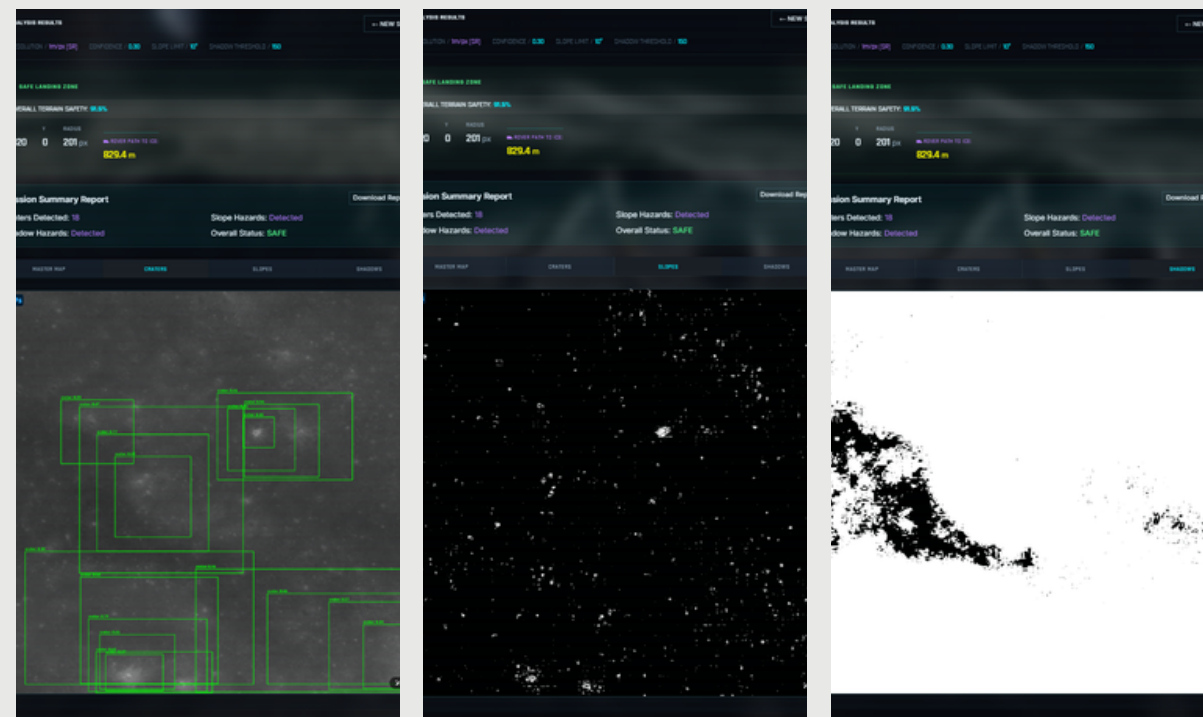


Fig: Lunar Hazard Mapper UI -Auto-detecting Safe Landing Zone (X,Y)

REFERENCE RESEARCH & DATASETS

- **ISRO Chandrayaan-2 TMC-2 Datasets:** ISSDC Pradan Portal —TMC-2 (Terrain Mapping Camera-2) 5m DEM & Ortho imagery used as core baseline.
- **ISRO OHRC Benchmarks:** Lunar Reconnaissance Orbiter Camera (0.5m/px) and Chandrayaan-2 OHRC (0.25m/px) data used for model validation.
- **Faster R-CNN Object Detection:** Ren, S., He, K., Girshick, R., & Sun, J. 'Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks', IEEE TPAMI.
- **Deep Super-Resolution Networks:** Wang, X., et al. 'ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks', ECCV Workshops.
- **Euclidean Distance Transform Algorithm:** Felzenszwalb, P., & Huttenlocher, D. 'Distance Transforms of Sampled Functions', Cornell Computing & Information Science.

HAZARD MAP COMPARISON & PROJECT STANDARDS

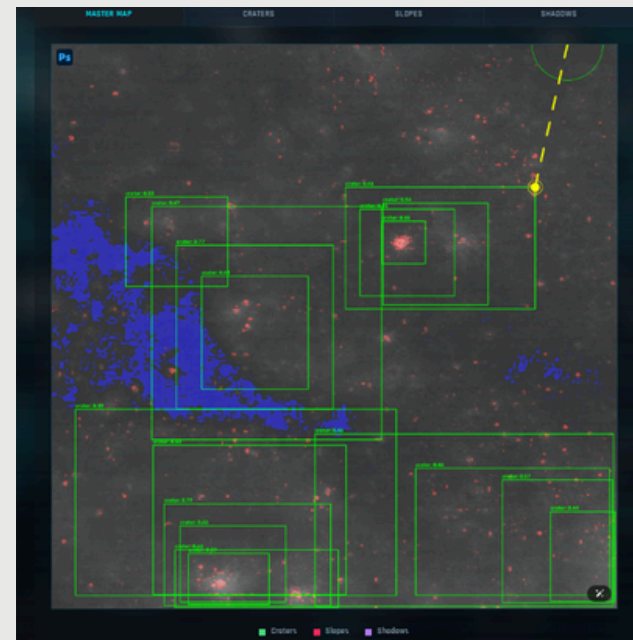


Fig: Hazard Map Comparison between non-enhanced TMC and AI Super-Resolved 1m Grid Map

- **Compliance:** Built strictly in accordance with ISRO Lander HDA Safety Tolerances (Slope $<10^\circ$, Boulder/Crater depth $<1\text{m}$).